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You just unboxed a barcode scanner.

Zebra ships two manuals with this thing. Together they run past 490 pages, and roughly 400 of those are walls of programming barcodes you will never scan. This is the part that matters, with the barcodes you actually need printed alongside. Scan them straight off the paper.

What is in the box

- The scanner, with the battery already fitted
- A Quick Start Guide
- **Nothing else.** The cradle, the USB cable and the power supply are all sold separately

Which manual, when

- **Quick Start Guide**, 22 pages. Cabling and the handful of setup barcodes.
- **Product Reference Guide**, 476 pages. Every setting that exists. Use it as a lookup, never as a read.

Sixty seconds to your first scan

- 1 **Charge it before anything else.** It arrives part charged. Empty to full is about 5 hours over standard USB, or 1 hour from an external 5V supply.
- 2 **Decide how it talks to your computer.** Cable into a USB port, or Bluetooth. Cable is the one that always works, so start there if you have the option.
- 3 **Open a text editor first.** Notepad, TextEdit, anything. A scanner is just a very fast keyboard, so you want somewhere for the typing to land.
- 4 **Pull the trigger at a barcode.** Aim the red line across the bars, not along them. About 15cm away is the sweet spot for a retail barcode.
- 5 **You should hear one beep and see a green flash**, and the digits should appear as text. That is the whole job. Everything after this page is refinement.

THE ONE THING WORTH LEARNING EARLY

You configure this scanner **by scanning barcodes out of a manual**. There is no settings screen. Each barcode you scan writes a setting into the scanner and it survives being switched off. That is why the reference guide is mostly barcodes, and it is why a printed page like this one can set the whole thing up.

Three ways to connect, in order of how much grief they cause

Pick one. You do not need the other two.

1. USB cable

Straight into the computer. No pairing, no battery anxiety, no dropouts. Boring and correct.

2. Bluetooth

The scanner pairs directly to a computer, tablet or phone as a wireless keyboard. This is what you want for stock taking.

3. Cradle

Sold separately. Charges the scanner and props it up so it scans automatically when you wave something at it.

Going in by cable

Plug it in and it will almost certainly just work, because USB keyboard is the factory setting. If it types nothing, scan this to put it back into keyboard mode.

ACT LIKE A USB KEYBOARD



USB Keyboard HID

Reference guide, page 8-5

Going in by Bluetooth

Scan the barcode below, then open Bluetooth settings on your computer or tablet and pair with the DS2278 the same way you would pair a keyboard.

PAIR AS A WIRELESS KEYBOARD



HID Bluetooth Classic

Reference guide, page 6-6

HOW YOU KNOW BLUETOOTH WORKED

A **low beep then a high beep, with a green LED**, means the link is up. Going the other way, **high then low with a red LED**, means it just dropped. A **long low then a long high** means it tried to reach the host and could not, so the other device is out of range, switched off, or not actually listening.

RANGE, HONESTLY

Bluetooth 4.0, Class 2. Zebra quotes a minimum of 10m in clear line of sight, and up to 100m only when it is paired with their cradle. Walls, shelving and metal racking all cut into that hard. Treat 10m through one wall as the realistic working figure.

If pairing will not take

- Delete any old DS2278 entry from the Bluetooth list first. A stale pairing blocks a new one.
- Scan **HID Bluetooth Classic** again, then pair within a minute or so before it stops advertising.
- A fast blinking green LED means it is trying to reconnect to something it already knows. Let it finish or clear the old pairing.
- Still nothing? Cable it in, confirm the scanner itself works, then come back to Bluetooth.

Switching between them later

Nothing is permanent. Scanning the barcode for one mode simply overrides the other, and factory defaults on page 9 returns it to USB keyboard.

Plugging the USB cable in while paired over Bluetooth does **not** automatically switch it across. If you cable it up and get nothing, scan **USB Keyboard HID**.

Aiming, and the two settings worth changing on day one

How to actually aim it

- The red line must cross the bars, **not run along them**. On a tall barcode, sweep across the middle.
- Do not hold it square on. A slight tilt kills the glare that stops a decode. It tolerates up to 65 degrees of skew and pitch, so angle is free.
- Roll does not matter at all. The scanner reads at any rotation.
- Closer is not better. Back off until the whole barcode sits inside the aiming line.
- Moving target? It keeps up to about 13cm per second in the hand, and 76cm per second when something is swept past it in the cradle.

When it will struggle

- **A dirty window.** The single most common cause of a scanner that has "got worse". Wipe it with a lens cloth.
- **Low contrast printing.** It needs 25 percent reflective difference. A faded thermal label is a genuine fail, not your technique.
- **Curved or crumpled labels**, particularly on bottles. Flatten what you can.
- **Screens.** It will read a barcode off a phone, but raise the phone brightness first.
- **Direct sun.** It copes to 107,600 lux, which is bright daylight, but glare on glossy labels still defeats it.

Trigger, or hands free

Out of the box you pull the trigger to scan. Presentation mode instead leaves it watching, so it fires the moment a barcode enters its view, which is what you want when it is sitting in a cradle and you are scanning with both hands full. The cost is battery.

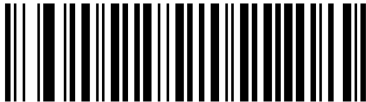
DEFAULT, PULL THE TRIGGER



Standard (Level)

Reference guide, page 5-16

HANDS FREE, ALWAYS WATCHING



Presentation (Blink)

Reference guide, page 5-16

Volume

It ships loud, which is right in a shop and wrong in an office. The beep is your only confirmation a scan worked, so turn it down rather than off.

DEFAULT, LOUD



High Volume

Reference guide, page 5-7

QUIETER



Low Volume

Reference guide, page 5-7

THE PROBLEM EVERYBODY HITS

It scans fine, but everything piles into one cell

You scan ten items into a spreadsheet and get one cell holding ten barcodes glued together. Nothing is broken. The scanner types the digits and then stops, because nobody told it to press Enter afterwards. Scan the barcode below once and it will move to the next row on its own, forever.

FIXES IT. SCAN THIS ONE.

**Add Enter Key (Carriage Return / Line Feed)**

Reference guide, page 5-31

Prefer to move **sideways** along a row instead of down a column? Use Tab instead. Scan one or the other, not both.

SIDEWAYS INSTEAD OF DOWN

**Tab Key**

Reference guide, page 5-31

ONLY IF THOSE TWO MISBEHAVE

**<DATA> <SUFFIX 1>**

Reference guide, page 5-34

TRY IT IN A TEXT EDITOR FIRST

Before wiring the scanner into real software, scan four things into Notepad. Four lines means the suffix is right and any remaining problem lives in the software, not the scanner. If your software wants a plain carriage return and is tripping over the line feed, the barcode on the left switches the scanner to sending its own configurable suffix instead.

IF THE LETTERS COME OUT WRONG

Digits fine but letters scrambled, or everything in capitals, means the scanner is set to the wrong country keyboard layout, or Caps Lock is on. The scanner emulates a keyboard, so it inherits the layout of the machine it is typing into. Keyboard layouts live in Appendix B of the reference guide, and it supports over 90 of them.

The scanner only talks to you in beeps. Here is the dictionary.

Listen rather than watch. The LED is on top and you will usually be looking at the barcode.

Normal scanning

YOU HEAR	LED	IT MEANS
One medium beep	Green flash	Decoded, and the data was sent. This is success.
Low, medium, high	Green	Powering up.
Nothing at all	Green solid	Presentation mode is on and it is watching for a barcode.
Four long low beeps	Red	Transmission error. The data was thrown away. Usually the wrong host type is selected.
Five long low beeps	Red	Conversion or format error.
Low, low, low, extra low	Red	Parity error.
Four short high beeps	Red, 4 seconds	Battery is low. Shown when you release the trigger.

While you are scanning configuration barcodes

YOU HEAR	LED	IT MEANS
Long low, then long high	Red	Refused. Wrong barcode, wrong order, or you scanned Cancel.
High, low	Green	It wants a number. Go and scan digits from Appendix G.
High, low, high, low	Green	Setting saved. You are done.

Bluetooth

YOU HEAR	LED	IT MEANS
Low, high	Green	Connected.
High, low	Red	Disconnected.
Long low, long high	Red	Could not reach the other device. Out of range or switched off.
Four long beeps, low high low high	None	The other device actively refused the connection.
High, low, high, low	Green	A pairing barcode was scanned.
Low, high, low, high	Red	Batch memory is full. It cannot store any more scans.

TWO THAT WILL MISLEAD YOU

A red LED never means the barcode was bad. It means the scanner read it perfectly and then could not deliver it, so look at the cable, the pairing or the software, never the label. And **Cancel beeps exactly like an error**, long low then long high with a red LED, because Zebra gave it the error sound. When you scan Cancel deliberately, that noise is success.

What it reads out of the box, and what it ignores

Barcode formats are called symbologies. The DS2278 is a 2D imager, so it reads flat barcodes and square ones equally well. Most types are already switched on and you will never touch this page.

On by default	
EAN-13, EAN-8	ON
UPC-A, UPC-E	ON
Code 128	ON
Code 39	ON
Code 93	ON
Codabar	ON
Interleaved 2 of 5	ON
QR Code, MicroQR	ON
Data Matrix	ON
PDF417	ON
Aztec	ON

Off by default	
Discrete 2 of 5	OFF
MSI	OFF
Code 11	OFF
Trioptic Code 39	OFF
Bookland EAN	OFF
Han Xin	OFF
MicroPDF417	OFF
These are off because they are rare and easy to misread. Turn one on only if you are certain your labels use it. Their enable barcodes are in Chapter 13.	

RETAIL TRANSLATION

Anything you buy in a shop is **EAN-13** in Europe or **UPC-A** in the United States. Both are on. If you are printing your own labels, use **Code 128**, which holds letters as well as digits and is also on. That covers almost every real use.

WORTH KNOWING BEFORE YOU SWITCH THINGS ON

Every symbology you enable is one more pattern the scanner tries to match against, which slightly raises the chance of a wrong read on a damaged label. Enabling everything is not a safe default, it is the opposite. Leave the list alone unless something genuinely fails to scan.

What the digits on a retail barcode actually are

Part of a 13 digit EAN	Example	What it is
Country prefix	539	Where the number was issued, not where the product was made
Manufacturer	1234	The company that registered the product
Product	56789	Which of their products this is
Check digit	7	Calculated from the other twelve. If a bar is misread the arithmetic fails and the scanner stays silent rather than handing you a wrong number, which is why a scan that beeps can be trusted

BATCH MODE

Walking away from the computer with it

Batch mode makes the scanner store scans in its own memory instead of sending them live, then dump them all at once when it gets back. This is how you count stock in a stockroom with no signal.

READ THIS BEFORE YOU SCAN ANYTHING BELOW

If you paired over Bluetooth as a keyboard, which is what page 2 set up, then **only Out of Range Batch Mode will work**. The other batch modes need the host to send a command back to the scanner telling it to dump, and a plain keyboard connection has no channel for that. They will be refused with the long low, long high error beep. This is verified behaviour, not a guess. Out of Range mode needs no commands at all, which is exactly why it works.

TURN BATCH ON

**Out of Range Batch Mode**

Reference guide, page 6-33

KEEP IT ON ACROSS RESTARTS

**Persistent Batch Enable**

Reference guide, page 6-34

BACK TO LIVE SCANNING

**Normal**

Reference guide, page 6-33

How a run actually goes

- 1 Scan the first two barcodes above, once, ever. This is setup, not a per run ritual. The third one is how you undo it afterwards.
- 2 Leave the computer or tablet on, with the cursor already sitting in the cell or field you want the data to land in.
- 3 Walk off. When you go out of Bluetooth range the link drops and the scanner quietly starts storing instead of sending.
- 4 Scan your stock. It sounds exactly the same as normal scanning.
- 5 Walk back into range. It reconnects and types everything it stored, in order, on its own.

How much it holds

30,720 bytes of storage, and each scan costs its own length plus 3 bytes. For 13 digit retail barcodes that works out at roughly **1,900 scans** before it is full.

How you know it is full

A distinct **low, high, low, high with a red LED**. This is the one sound you must recognise, because a stored scan otherwise sounds identical to a sent one.

DO NOT COMBINE THIS WITH AUTO RECONNECT ON BARCODE DATA

There is a setting that tells the scanner to reconnect when you scan something. Turned on together with batch mode, the scan goes into the buffer instead of triggering the reconnect, so the scanner can never get back to the host. It hangs, permanently, and the only way out is a reset.

Start from the symptom, not the settings

No red aiming line when you pull the trigger

- Flat battery. Charge it for ten minutes and try again before assuming anything worse.
- On a cabled setup, the cable is loose or it is the wrong cable. Reseat both ends.
- If it is in a cradle, the cradle may need its power supply connected.

It beeps and flashes green, but nothing appears on screen

The scan worked and the delivery failed. In order of likelihood:

- Nothing is focused. Click into the field first. The scanner types, it cannot choose where.
- Bluetooth has silently dropped. Look for the green LED, or listen for the reconnect beeps.
- It is set to the wrong host type. Scan **USB Keyboard HID** from page 2.
- It is in batch mode and storing rather than sending. Scan **Normal** from page 7.

Every configuration barcode gets refused, long low then long high

- You may be halfway through a sequence that expects a number. Scan **Cancel** on page 9 to back out, then start again.
- The setting may genuinely not apply to how it is connected. Batch mode over Bluetooth keyboard is the classic example, covered on page 7.
- Check you are scanning from the DS2278 manual. Barcodes from another Zebra model will be rejected.

The data arrives but it is wrong

- Letters scrambled or everything uppercase means a keyboard layout mismatch or Caps Lock. See page 4.
- Everything on one line means no Enter suffix. See page 4.
- An extra digit on the end is the check digit, which it transmits by default. That is normal and usually wanted.
- Four long low beeps with a red LED means it was configured for a different host type entirely.

It used to scan well and now it hesitates

- Clean the window. This is the cause far more often than anything else.
- Use a lens cloth or a cloth dampened with mild dish soap or hydrogen peroxide. Never spray liquid into the window.
- **No isopropyl alcohol, no ammonia, no acetone, no bleach.** Zebra lists all of these as damaging to the plastics.

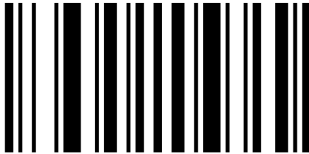
BEFORE YOU CALL ANYBODY

There is a **Report Software Version** barcode on page 4-7 of the reference guide. Scan it into a text editor and it types out the firmware version, which is the first thing support will ask for.

The two barcodes that undo everything else

If you have scanned your way into a corner, and everybody does at least once, these are the way back. Keep this page findable.

WIPES EVERY SETTING



Set Factory Defaults

Reference guide, page 5-5

Puts every setting back to how it left the factory. Your suffix, volume, batch mode and host type all go. It does **not** wipe firmware, and it does not physically unpair the scanner from a computer, so you may need to remove it from the Bluetooth list on that machine too.

BACKS OUT OF A HALF FINISHED SEQUENCE



Cancel

Reference guide, Appendix G-3

Some settings need a number scanned after them. If you start one and change your mind, the scanner sits there waiting and refuses everything else. Cancel gets you out. **It answers with the error beep on purpose.** That is it working.

WHERE THE NUMBERS LIVE

When a setting asks for a value, you enter it by scanning digits from the numeric barcode pages, **Appendix G** of the reference guide, pages G-1 and G-2. There is no keypad on the scanner. This is also why a stray scan can leave it waiting mid sequence.

Battery and charging, briefly

THING	FIGURE
Battery	2,400 mAh lithium ion, and it is replaceable by hand
Scans per charge	110,000 at a steady 60 scans a minute
Working time	Around 84 hours of realistic checkout use
Empty to full	5 hours on standard USB, 1 hour from an external 5V supply
Charging	Solid green while charging in a cradle. Fast blinking green on micro USB, solid green when full
Solid red	Charging fault. Reseat it, and try a different supply before assuming the battery is dead

Where to look when this sheet runs out

Page numbers below are Zebra's own printed numbers, the ones like 5-34 shown at the top of each page, not the PDF page count. Chapters restart their numbering, so searching for the printed number is faster than scrolling.

Product Reference Guide

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Symbologies	Ch 13
Keyboard countries	App B
Numeric barcodes	App G

The numbers that matter

Drop rating	1.5m onto concrete, repeatedly
Sealing	IP52. Dust protected, drips only
Operating range	0 to 50 degrees C
Weight	214g
Bluetooth	4.0 with BLE, Class 2

Three habits worth keeping

- **Test into a text editor first**, every time you change a setting. It separates a scanner problem from a software problem in seconds.
- **Change one setting at a time**. Scanning four configuration barcodes in a row and then testing tells you nothing about which one did what.
- **Keep this page**. Factory defaults on page 9 undoes anything you can do to it, so nothing here is unrecoverable.

IF YOU OUTGROW SCANNING BARCODES OFF PAPER

Zebra publish a free Windows tool called **123Scan**. Plug the scanner in by USB and you get a normal settings window, plus the ability to save a configuration and apply it to more scanners later. For one scanner and a handful of settings, the barcodes in this document are quicker. For ten scanners set up identically, it is the better route.

Your first week, in order

- 1 **Charge it fully** before judging anything about battery life.
- 2 **Scan into a text editor** and confirm you get clean digits. Page 1.
- 3 **Add the Enter key**. This is the one setting almost everybody needs. Page 4.
- 4 **Set the volume** to suit the room, rather than turning the beep off. Page 3.
- 5 **Scan the ugliest labels you own**, the creased and faded ones, while you can still return the scanner.
- 6 **Learn the full memory sound** before you rely on batch mode for a real stock count. Page 7.